

ChE 210 – Material and Energy Balance –Fall 2022

Student Learning Outcome 1

Rubric for assessment problem on final exam (Problem #2)

SLO 1 Performance Indicator (PI)	(3) Satisfactory	(2) Developing	(1) Unsatisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject	Able to successfully combine mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development	Does not understand the connection between mathematical models and the system or process to be analyzed or designed
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra	applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems
4. Translates academic theory into engineering applications	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality	Some gaps in understanding the application of theory to the problem and expects theory to predict reality	Does not appear to grasp the connection between theory and the problem

Averages:

Performance Indicators: (1) 1.95 (2) 1.79 (4) 2.58 Overall Average: 2.11

PI4 is the strongest score. This score was evaluated on students' ability to apply the degree of freedom analysis to the material balance on a mixing process. The students who struggle with this task use a rote approach to applying the DOF analysis. To improve performance and understanding, many examples are worked in class demonstrating the relation between the information given in the problem with the values in the DOF. Also, the DOF is repeated after the equations are set up by simply counting unknowns and independent equations.

The next highest score is for PI1 which corresponds to being able to set up the material balance equations for the chosen system. For this example, students needed to identify the input and output stream variable for the four species being processed in the mixer. About half the class was successful while the other half had unsatisfactory results. This is a new skill for many students in this course which has still not developed by the time of the first exam. Students who do not work the homework assignments independently tend to take longer in developing this skill. There is some resistance from these students to working actively to acquire this skill. By emphasizing the importance of being able to develop model equations in future chemical engineering courses and in future industrial applications, students can be motivated to engage the topic with more focus and ultimately success.

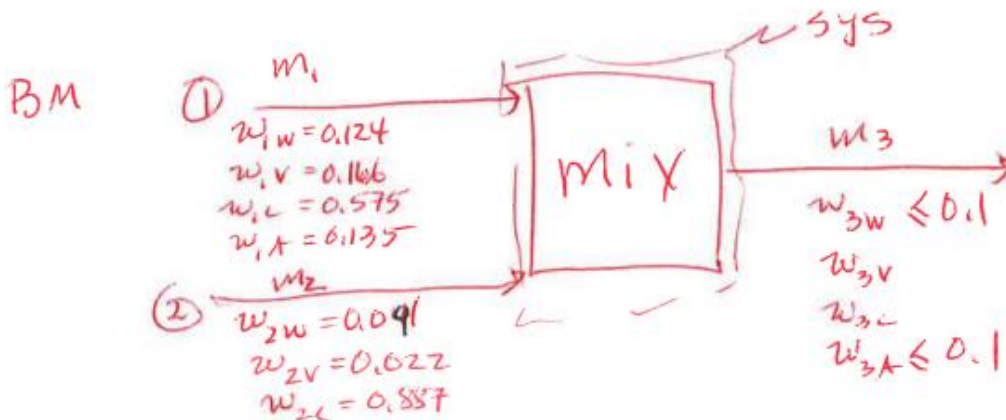
Finally, the PI2 is the lowest score. This is perhaps due to the fact that failure to set up the equations made solving them impossible. When we look at the success of students who were able to develop the model equations correctly, is much higher. A significant percentage of students arrive in ChE 210 with inadequate

math skills, even at the level of solving systems of algebraic equations. Solving linear and non-linear algebraic equations is a prerequisite skill for the course; however, students' proficiency has decreased recently. More time and attention needs to be focused on the deficiency. The course does cover numerical methods of solving such equations. Lecture examples that go through more detailed analytical methods should be developed and presented. Application of the degree of freedom analysis can guide students to finding solution strategies. There is a required discussion section where students' work examples during the discussion period and have their work reviewed at the time. They can also have their questions addressed during discussion or during office hours with the TA or instructor.

NAME

Solution

- (2) (50 pts.) Charcoal is to be manufactured that is specified not to contain over 10 wt% water nor over 10 wt% ash. The base material used in the process has the following composition: 12.4 wt% water, 16.6 wt% volatiles, 57.5 wt% carbon, and 13.5 wt% ash. To meet both specifications, the base material will be mixed with petroleum coke which has the composition: 9.1 wt% water, 2.2 wt% volatiles, and 88.7 wt% carbon. Use the minimum amount of petroleum coke necessary to meet the specifications. Assume steady operation.
- (a) Draw a block flow diagram of the process and label all the streams. Define your system. (5 pts.)



- (b) Do a degree of freedom analysis following the method on page 105 of the textbook by Murphy. In the table below, give values for 4a, 4b, 5a, 5b, 5c, and 5d along with a brief explanation for the value you give. Compute the DOF. (10 pts.)

	Value	Explanation
4a. Stream Variables	11	4 in ① 3 in ② 4 in ③
4b. System Variable	0	no chem Rx no accum
Total Independent Variables	11	11 + 0
5a. Specified Flows	0 → 1	need a basis $m_1 = 100 \text{ kg}$
5b. Specified Stream Compositions	5	3 for ① and 2 for ②
5c. Specified System Performances	2	$w_{3w} \leq 0.1$ $w_{3A} \leq 0.1$
5d. Material Balance Equations	4	w, v, c, A
Total Independent Equations	12	1 + 5 + 2 + 4
DOF	-1	11 - 12

cannot pick values for both w_{3w} and w_{3A} !

- (c) Calculate the appropriate mass ratio of base material to petroleum coke to satisfy both specifications **using the minimum amount of petroleum coke per mass of the base material**. Calculate the composition of the product in wt%. Fill in your answers in the spaces on the back side of this sheet. (35 pts.)

To get min m_2 w_{3w} or w_{3A} must equal 0.1

Species mass balances ($m_1 = 100$ kg Basis)

$$W \quad 12.4 \text{ kg}_W + 0.091 m_2 = w_{3W} m_3$$

$$V \quad 16.6 \text{ kg}_V + 0.022 m_2 = w_{3V} m_3$$

$$C \quad 57.5 \text{ kg}_C + 0.887 m_2 = w_{3C} m_3$$

$$A \quad 13.5 \text{ kg}_A = w_{3A} m_3$$

5 eqns

6 unknowns:

$$m_2, m_3, w_{3W}, w_{3V}, w_{3C}, w_{3A}$$

can pick either

$$w_{3W} = 0.1 \text{ or } w_{3A} = 0.1$$

$$\text{constraint: } w_{3W} + w_{3V} + w_{3C} + w_{3A} = 1$$

$$\text{Total mass } 100 \text{ kg} + m_2 = m_3$$

$$\text{Try } w_{3A} = 0.1 \frac{\text{kg}_A}{\text{kg}} \Rightarrow m_3 = \frac{13.5 \text{ kg}_A}{0.1 \text{ kg}_A/\text{kg}} = 135 \text{ kg} \Rightarrow m_2 = 35 \text{ kg}$$

$$W: w_{3W} = \frac{12.4 \text{ kg}_W + 0.091(35)}{135} = 0.115 \frac{\text{kg}_W}{\text{kg}} > 0.1 \text{ violates constraint}$$

$$\text{Try } w_{3W} = 0.1 \frac{\text{kg}_W}{\text{kg}}$$

$$W: 12.4 + 0.091 m_2 = 0.1 (100 + m_2)$$

$$12.4 - 10 = (0.1 - 0.091) m_2$$

$$m_2 = \frac{2.4}{0.009} \text{ kg} = 266.6 \text{ kg}$$

$$m_3 = 366.6 \text{ kg}$$

$$A: w_{3A} = \frac{13.5}{366.6} = 0.0368 \frac{\text{kg}_A}{\text{kg}} \leq 0.1 \text{ satisfies constraint}$$

$$V: w_{3V} = \frac{16.6 + 0.022(266.6)}{366.6} = 0.0613 \frac{\text{kg}_V}{\text{kg}}$$

$$w_{3C} = 1 - w_{3W} - w_{3A} - w_{3V} = 0.8019$$

$$\text{Mass of base material/mass of petroleum coke} = \frac{100}{266.6} = 0.375 \text{ kg BM/kg PC}$$

Product Composition:

$$\text{Water wt\%} = 10 \quad \text{Volatiles wt\%} = 6.13 \quad \text{Carbon wt\%} = 80.19 \quad \text{Ash wt\%} = 3.68$$